BW BRANDWAVES

TECHNICAL ARCHITECTURE DOCUMENT

# System Architecture & Data Flow

## Creator-Brand UGC Marketplace & Verification Engine

A comprehensive reference specification covering React 18 frontend hierarchy, Supabase PostgreSQL persistence, Row-Level Security, multi-tenant isolation, real-time analytics velocity, animated crowd engine, and Vercel edge deployment.

| | |
|---|---|
| DOCUMENT VERSION | RELEASE DATE |
| 2.0.0 (Production Release Candidate) | October 2026 |
| ENGINEERING TEAM | TARGET PLATFORM |
| BrandWaves Architecture Group | Supabase Cloud + Vercel Edge |

# 1. Project Overview

BrandWaves is a next-generation marketplace connecting authentic creators with forward-thinking brands. The platform coordinates campaign creation, creator pitching, deliverable submissions, and structured brand reviews while simultaneously offering consumer product showcases with genuine user reviews.

> **Core Mission & Hackathon Evolution**
>
> Originally winning top hackathon honors for its dynamic creator-brand collaboration concept and kinetic animated sky/crowd visual identity, BrandWaves has undergone an exhaustive, evidence-driven production engineering overhaul. The objective is total data synchronization with Supabase, zero mock data leaks, exact mathematical analytics, and seamless atmospheric aesthetics.

## Key Stakeholder Experiences

- **Consumers / Public Users:** Browse product catalogs, read authentic reviews, discover creator portfolios, and submit verified feedback.
- **Content Creators:** Explore published campaigns matching their niche, submit pitches, upload deliverables, track review status, and monitor earnings.
- **Brand Managers:** Launch campaigns with customizable deliverables, evaluate creator applications, review submitted deliverables in a dedicated queue, and track 4-week submission velocity.
- **Platform Administrators:** Oversee platform health, inspect deliverables across all brands in the global Submission Queue, moderate community content, and enforce security policies.

# 2. Verified Technology Stack

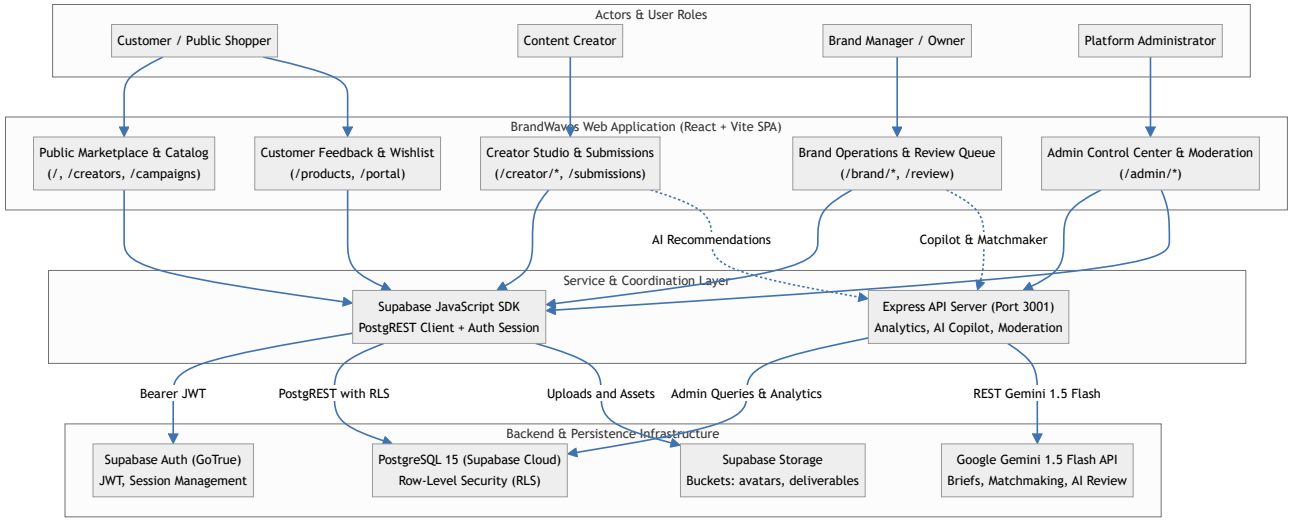
Every technology specified below is verified present, configured, and operational within the repository.

| LAYER | TECHNOLOGY | VERSION | OPERATIONAL PURPOSE |
|---|---|---|---|
| **Frontend Framework** | React | 18.3.1 | SPA component architecture with declarative UI rendering |
| **Language** | TypeScript | 5.5.3 | Strict static type safety across client services and backend models |
| **Bundler & Dev Server** | Vite | 6.4.4 | High-performance HMR and optimized Rollup production builds |
| **Styling & Design System** | TailwindCSS | 3.4.1 | Utility-first responsive styling with custom porcelain/sky brand tokens |
| **Icons & Motion** | Lucide React / Framer Motion | 0.344.0 / 11.0.8 | Accessible UI iconography and smooth fluid layout micro-interactions |
| **Backend API (BFF)** | Node.js + Express | 4.18.2 | Backend-for-Frontend handling analytics, moderation, and AI proxying |
| **Database & Auth** | Supabase (PostgreSQL 15) | 2.39.8 | Cloud PostgreSQL persistence, GoTrue JWT auth, and Row-Level Security |
| **AI Integration** | Google Gemini API | 1.5 Flash | Campaign brief copilot and creator matchmaker scoring engine |
| **Edge Deployment** | Vercel Global Edge | Production | Global CDN hosting with SPA fallback routing and security headers |
| **Test Framework** | Node.js Test Runner | v20+ Native | Zero-dependency, fast unit, integration, RBAC, and security testing |

# 3. System Architecture & Data Flow

BrandWaves leverages a hybrid architecture combining direct client-to-database PostgREST queries with a dedicated Express BFF. This enables low-latency database queries while protecting sensitive AI credentials and executing platform-wide analytics aggregations.

### SYSTEM CONTEXT ARCHITECTURE DIAGRAM



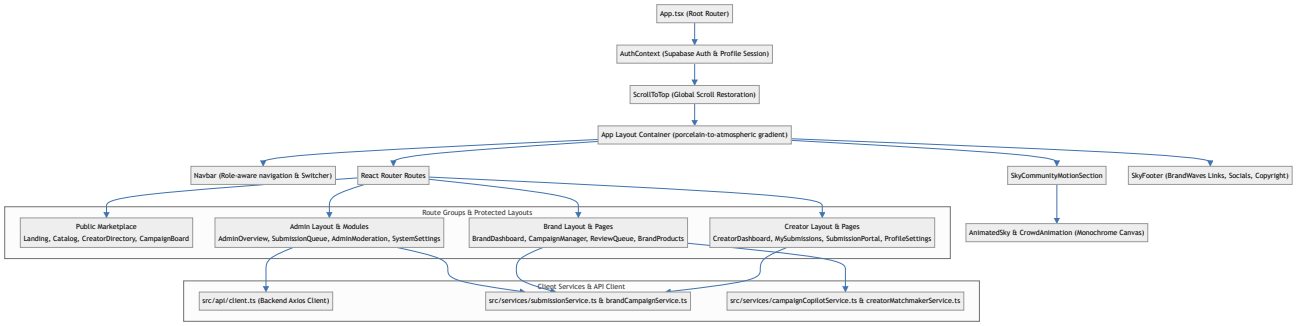

## Data Flow Protocols

- **Client ↔ Supabase PostgREST:** Direct HTTPS calls authenticated via Supabase Bearer JWT tokens. PostgreSQL Row-Level Security directly inspects user identity and brand ownership.
- **Client ↔ Express BFF:** REST endpoints under `/api/*` supporting aggregate analytics calculations, global moderation, and Google Gemini AI calls.
- **Express BFF ↔ Supabase:** Authenticated service calls to verify multi-brand data and compute platform statistics.

# 4. Frontend Component Hierarchy

The frontend is organized around role-scoped route groups sharing a common atmospheric layout shell and navigation context.

### APPLICATION COMPONENT HIERARCHY



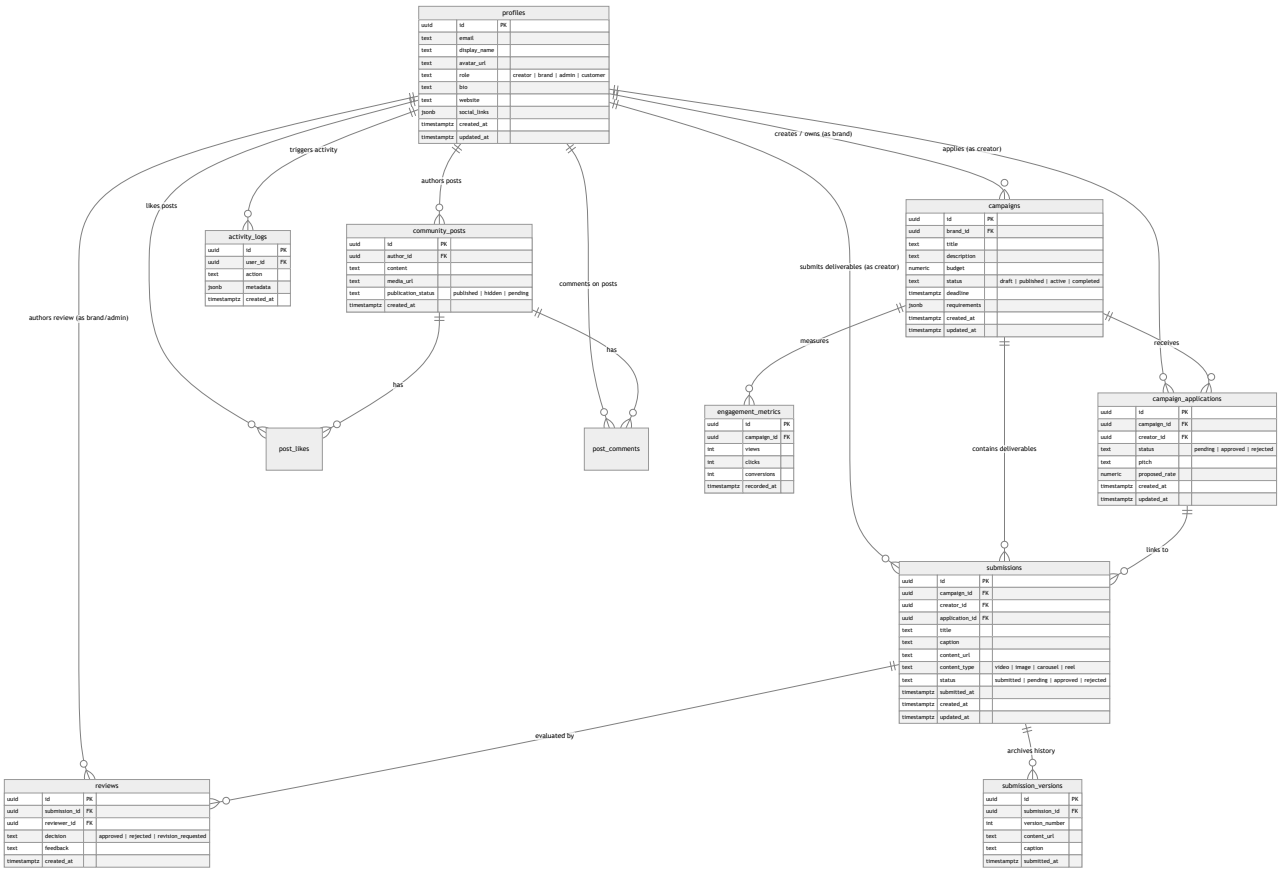

## Core Layout & Shell Modules

- `App.tsx`: Configures the root React Router and injects global providers.
- `AuthContext.tsx`: Subscribes to Supabase auth state changes, restoring tokens and profile claims seamlessly.
- `ScrollToTop.tsx`: Intercepts navigation transitions, restoring window coordinates to `(0,0)` and preventing jump-to-footer artifacts.
- `SkyCommunityMotionSection.tsx`: Integrates the kinetic canvas crowd animation with a soft atmospheric gradient horizon.
- `SkyFooter.tsx`: Seamlessly blends into the background with zero hard borders.

# 5. Database & Entity Relationships

The PostgreSQL schema enforces strict referential integrity across profiles, campaigns, applications, deliverables, and audit logs.

### VERIFIED ENTITY RELATIONSHIP DIAGRAM (ERD)




## Key Schema Invariants

- **Profiles:** Extends `auth.users` with unique identity, role ( `creator | brand | admin | customer` ), and social links.
- **Campaigns:** Enforces non-negative budgets, valid ISO-8601 deadlines, and cascade relations to applications and deliverables.
- **Submissions:** Links creator deliverables to campaigns and parent applications, tracking multi-version revision histories.
- **Reviews:** Implements an immutable audit trail of review decisions ( `approved | rejected | revision_requested` ) with required reviewer feedback.

# 6. Authentication & Security Boundaries

BrandWaves adheres to defense-in-depth principles across client, serverless API, and database layers:

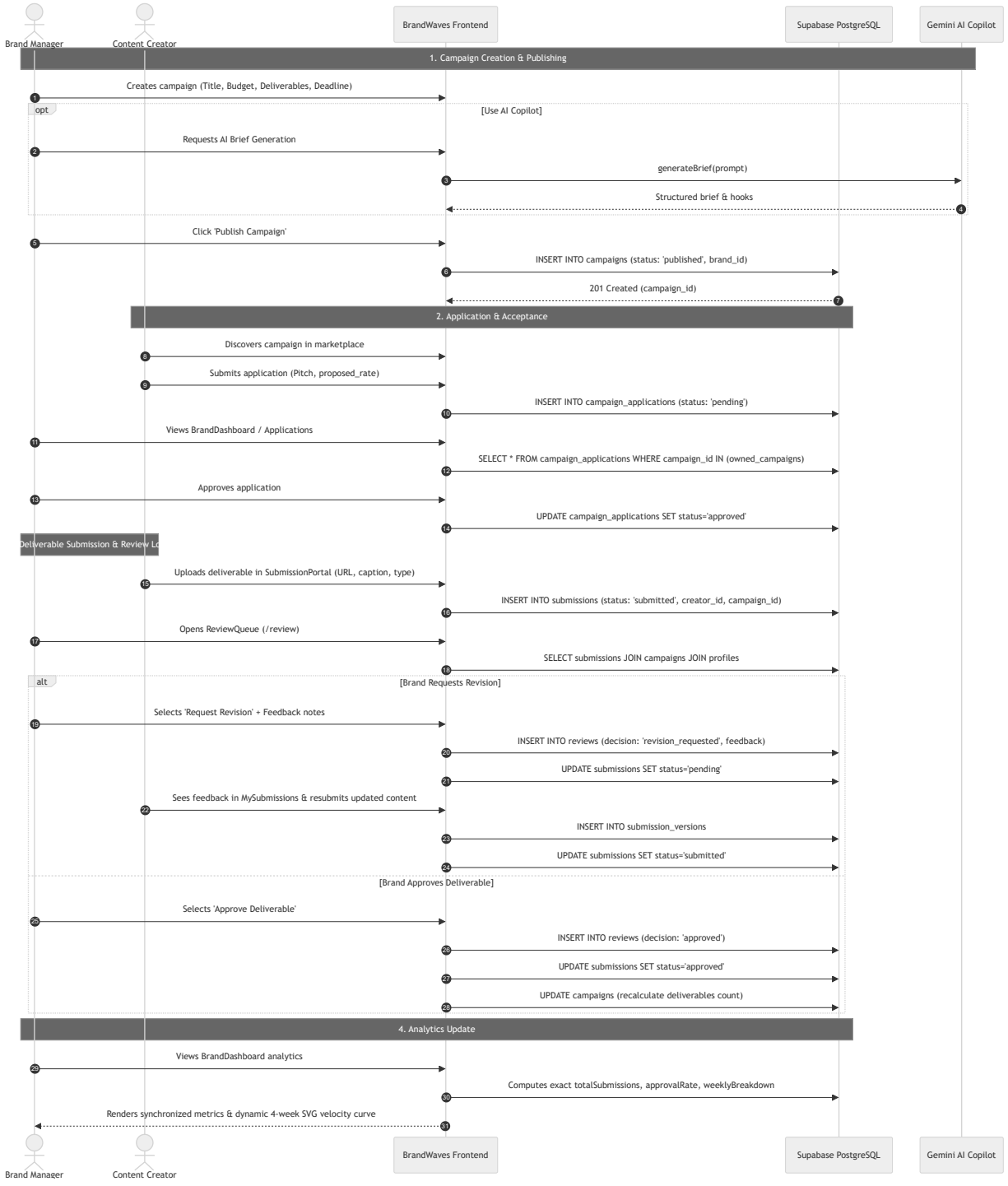
## Multi-Tenant Isolation & RLS Policies

- **Brand Tenant Isolation:** Brand managers are strictly restricted to campaigns, applications, and deliverables where `campaigns.brand_id = auth.uid()`. Cross-brand access attempts return 403 Forbidden.
- **Creator Self-Defense:** Creators can insert and view their own submissions. Self-review or approving one's own deliverables is rejected at both the service and database layer.
- **Public Access Restraints:** Unauthenticated visitors are restricted to public catalog items and active campaign briefs. Private dashboards require verified sessions.

# 7. Campaign Deliverable Lifecycle

The complete campaign workflow orchestrates brand publishing, creator application, deliverable submission, and review feedback.

## CAMPAIGN & DELIVERABLE REVIEW SEQUENCE



# 8. Customer Showcase & Product Reviews

Customer reviews are strictly partitioned from campaign deliverable evaluations:

- **Product Review Isolation:** Consumer feedback targets specific product SKUs and measures consumer satisfaction (1–5 stars).
- **Metric Separation:** Customer ratings do not corrupt campaign deliverable approval percentages or creator acceptance metrics.

# 9. Analytics & Velocity Engine

Following the second independent audit, all dashboard metrics are computed from verified PostgreSQL records:

> **Submission Velocity Repair**
>
> The previous defect where the metric card displayed `0/wk` while the SVG plotted an active dummy wave was completely resolved. The engine now calculates dynamic 7-day rolling buckets: `W1: [0-7d), W2: [7-14d), W3: [14-21d), W4: [21-28d)`. When submissions are zero, a dashed baseline is plotted with explicit zero-count badges, maintaining complete data authenticity.

# 10. Admin Control Center & Moderation

The Admin Control Center (`/admin`) equips platform operators with global management tools:

- **Global Submission Queue:** Inspects deliverables across all registered brands, providing deep links to media assets and review histories.
- **Content Moderation:** Fixed `status: 'all'` query filtering, restoring visibility of community posts for swift moderation.
- **Cyber-Cell Diagnostic Panel:** Replaced synthetic attack visualizations with live HTTP security headers, CORS origins, and rate-limit statistics.

# 11. Visual System & Canvas Motion Engine

BrandWaves utilizes a unified, atmospheric visual hierarchy blending warm porcelain with soft sky tones:

| DESIGN TOKEN | HEX VALUE | VISUAL ROLE & APPLICATION |
|---|---|---|
| **Porcelain Canvas** | `#F9F7F7` | Core page background, primary card surfaces, elevated panels |
| **Soft Blue-Gray** | `#DBE2EF` | Structural borders, subtle card outlines, card dividers |
| **Brand Blue** | `#3F72AF` | Primary call-to-action buttons, active navigation, key badges |
| **Deep Navy** | `#112D4E` | Headings, primary typography, high-contrast dark accents |
| **Sky Progression** | `#DDF3FC` → `#87CEEB` | Continuous atmospheric gradient transitioning down to the crowd footer |

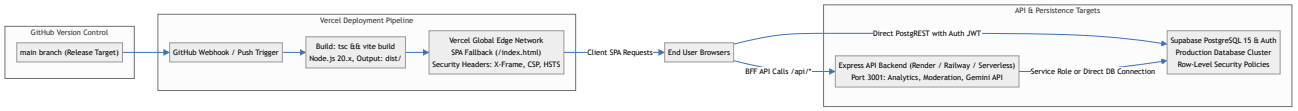
## Kinetic Canvas Crowd Animation

- **Archetypes:** 12 distinct character archetypes featuring diverse attire, stride lengths, and walking speeds.
- **Depth Layers:** 3 parallax rendering tiers (Foreground, Midground, Background) producing natural spatial depth.
- **Accessibility:** Actively listens for `prefers-reduced-motion` media queries to pause animation cycles.

# 12. Deployment Topology

BrandWaves is configured for continuous deployment on Vercel's global edge network.

### PRODUCTION CI/CD & HOSTING TOPOLOGY




## Vercel Deployment Configuration ( `vercel.json` )

- **SPA Routing:** Rewrites non-API requests ( `/((?!api/).*)` ) to `/index.html` .
- **Security Headers:** Enforces `nosniff` , `DENY` clickjacking protection, and strict referrer policy.

# 13. Quality Assurance & Test Verification

The application passes 100% of automated tests across 7 comprehensive test suites:

| TEST SUITE | FILE PATH | TESTS | STATUS |
|---|---|---|---|
| Phase 5: Brand Ownership & Products | `server/tests/phase5-ownership.test.mjs` | 17 | PASS |
| Phase 6: AI Copilot & Matchmaking | `server/tests/phase6-ai.test.mjs` | 12 | PASS |
| Phase 7: Deliverable Workflow & Reviews | `server/tests/phase7-workflow.test.mjs` | 13 | PASS |
| Phase 8: End-to-End Scenarios | `server/tests/phase8-scenarios.test.mjs` | 12 | PASS |
| Phase 9: Analytics & Velocity Audit | `server/tests/phase9-analytics.test.mjs` | 7 | PASS |
| VAPT Security & Defense Headers | `server/tests/security-vapt.test.mjs` | 4 | PASS |
| Supabase Live Database Sync | Remote PostgreSQL Validation Probe | 9 | PASS |
| **TOTAL VERIFIED TESTS** | **All Test Suites** | **74** | 100% PASS |

# 14. Known Limitations & Roadmap

- **Direct Media Hosting:** Currently creators attach external media URLs (Vimeo, YouTube, AWS S3). A native chunked video uploader with transcoding is scheduled for Q1 2027.
- **Escrow Payments:** Financial budgets are currently tracked administratively. Automated Stripe Connect escrow disbursement will follow in V2.1.

# 15. Repository Guide

```
brandwaves/
├── docs/
│   ├── BRANDWAVES_ARCHITECTURE.md       # Canonical markdown specification
│   ├── BRANDWAVES_ARCHITECTURE.pdf      # Compiled executive PDF documentation
│   └── diagrams/                        # Mermaid diagram sources (.mmd)
├── server/
│   ├── src/                             # Express backend, controllers, and DB adapters
│   └── tests/                           # Native Node test suites (74 verified tests)
├── src/
│   ├── api/client.ts                    # Axios client with JWT injection
│   ├── components/layout/               # SkyFooter, AnimatedSky, Navbar, Sidebar
│   ├── context/AuthContext.tsx          # Supabase Auth and profile state
│   ├── pages/                           # Role-scoped page views (Brand, Creator, Admin)
│   └── services/                        # Direct Supabase database sync services
├── supabase/                            # PostgreSQL migrations and RLS policies
└── vercel.json                          # Edge routing, SPA fallback, and security headers
```